



Commission for
Communications Regulation

FROM MINUTES TO MEGABYTES: IRELAND'S TELECOM TRANSFORMATION

GROUP 6

Project Problem Statement:

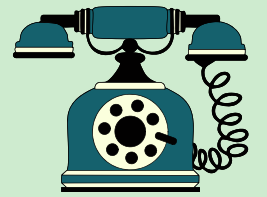
The Imbalance Between Connection and Conversion

Despite rapid subscriber growth and technological innovation, Ireland's telecom industry faces a growing imbalance: **the expansion of mobile and broadband services has not translated into proportional revenue growth.**

Our project investigates the root of this mismatch, exploring how technological evolution, changing consumer behaviour, and competitive market dynamics have reshaped the nation's communication landscape.

Everyone's online, but the profits didn't log in, that's the mystery we're untangling.

Fixed Voice



Mobile



Broadband



Data Selection: Capturing Ireland's Telecom Evolution (2015-2024)

From regulation to real insight, visualising Ireland's decade of digital transformation

Data obtained from the Irish Commission for Communications Regulation (ComReg), the official government body overseeing Ireland's electronic communications sector. Covering the years 2015 to 2024, and focuses on critical areas such as fixed voice, mobile, and broadband services to provide a comprehensive overview of the telecom landscape.

Research centered on three key service pillars:

Fixed Voice – legacy communication networks

Mobile – driver of mass connectivity

Broadband – backbone of Ireland's digital economy



Data Transformation: From Raw Tables to Actionable Insights

1

Step 1: Verifying and Reviewing Relevance

Exclude 2005-2014 due to inconsistencies and 2025 for incompleteness, focusing analysis on reliable 2015-2024 data.

2

Step 2: Data Segregation and Structuring

Segregated data by metric type and transposed Quarter-Year structure for Tableau compatibility.

3

Step 3: Naming Convention Standardisation

Renamed all datasets and column headers with consistent names to maintain uniformity across metrics and services.

4

Step 4: Handling Missing and Incomplete Data

Addressed data gaps by preserving missing values for new technologies like 5G and applying mean imputation for minor gaps.

5

Step 5: Key Creation and Data Integration

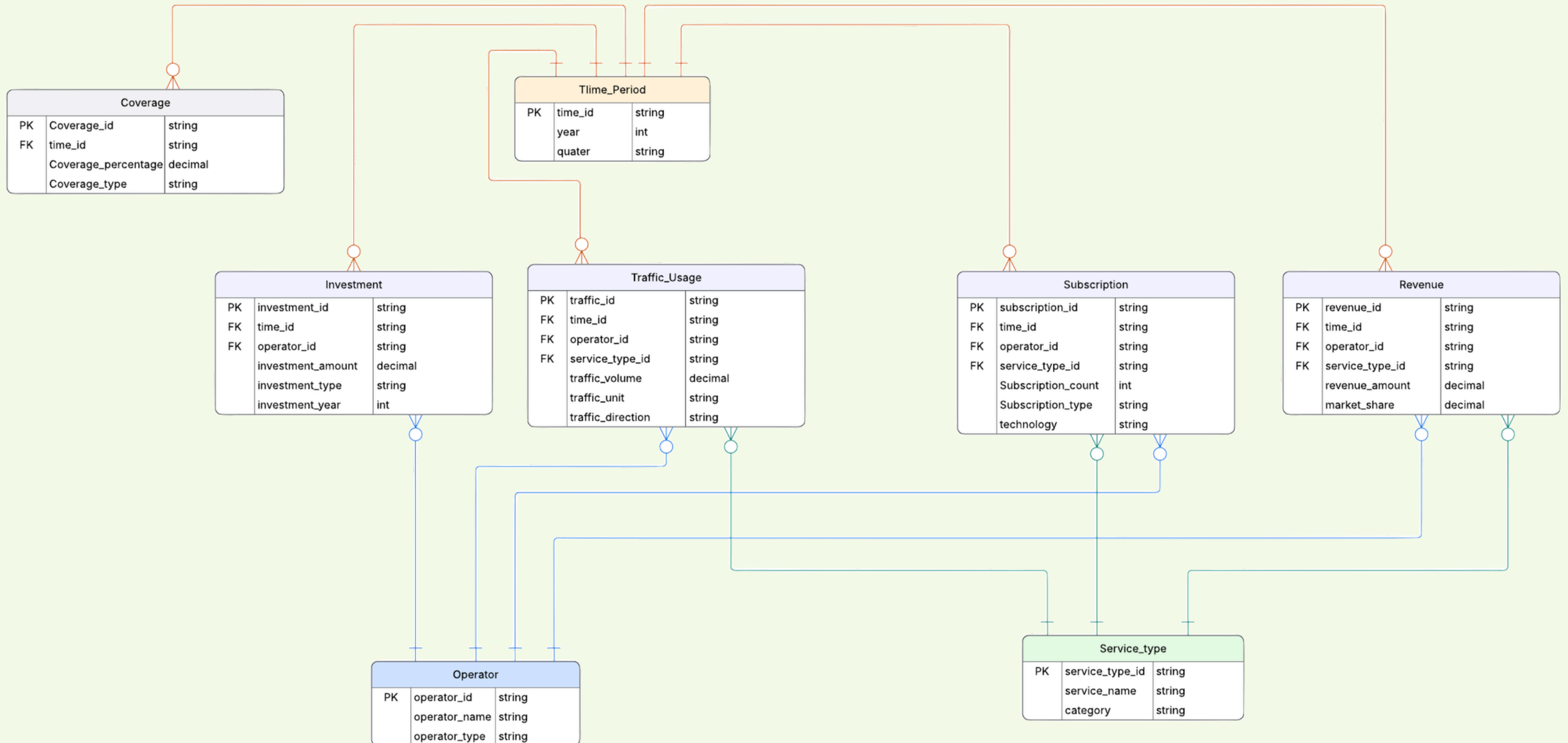
Implemented Time_ID as a foreign key across all datasets, with unique identifiers ensuring record uniqueness and integration.

6

Step 6: Final Validation and Quality Check

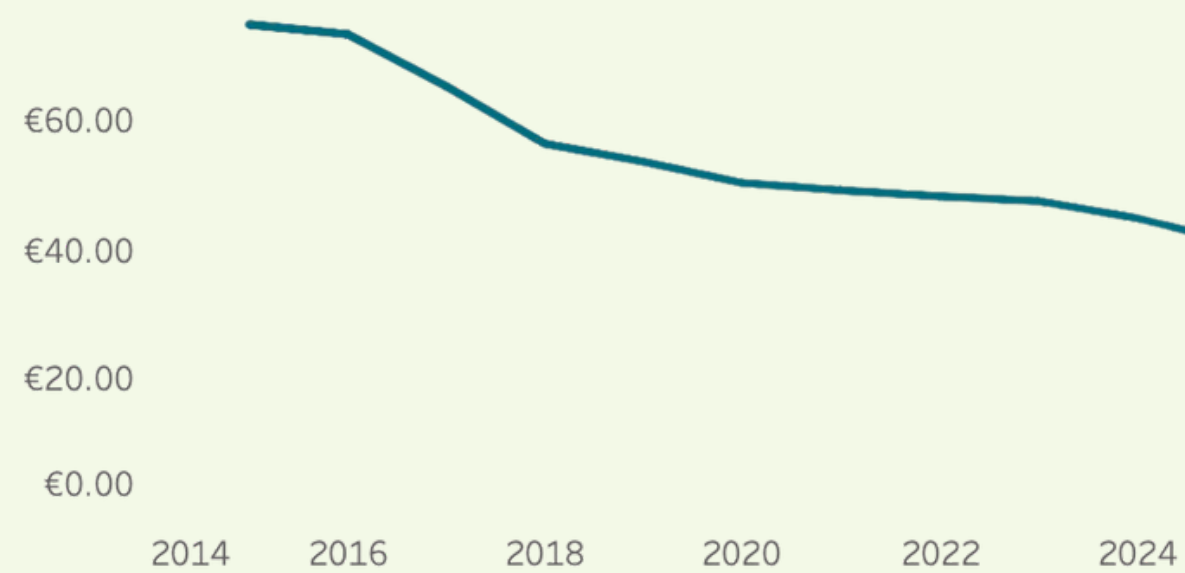
Performed data validation, added metric/segment columns, and ensured a clean dataset for accurate Tableau visualization.

The Logical Database Model



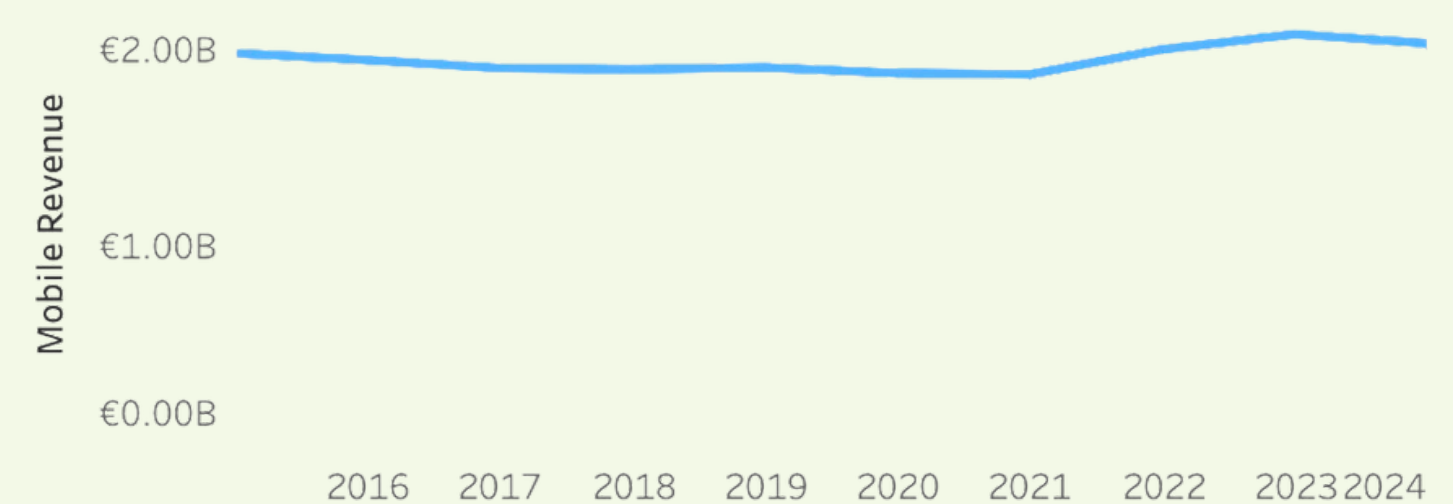
Trading ARPU for Volume

Mobile ARPU (per month)



- * ARPU saw a steady decline.

Mobile



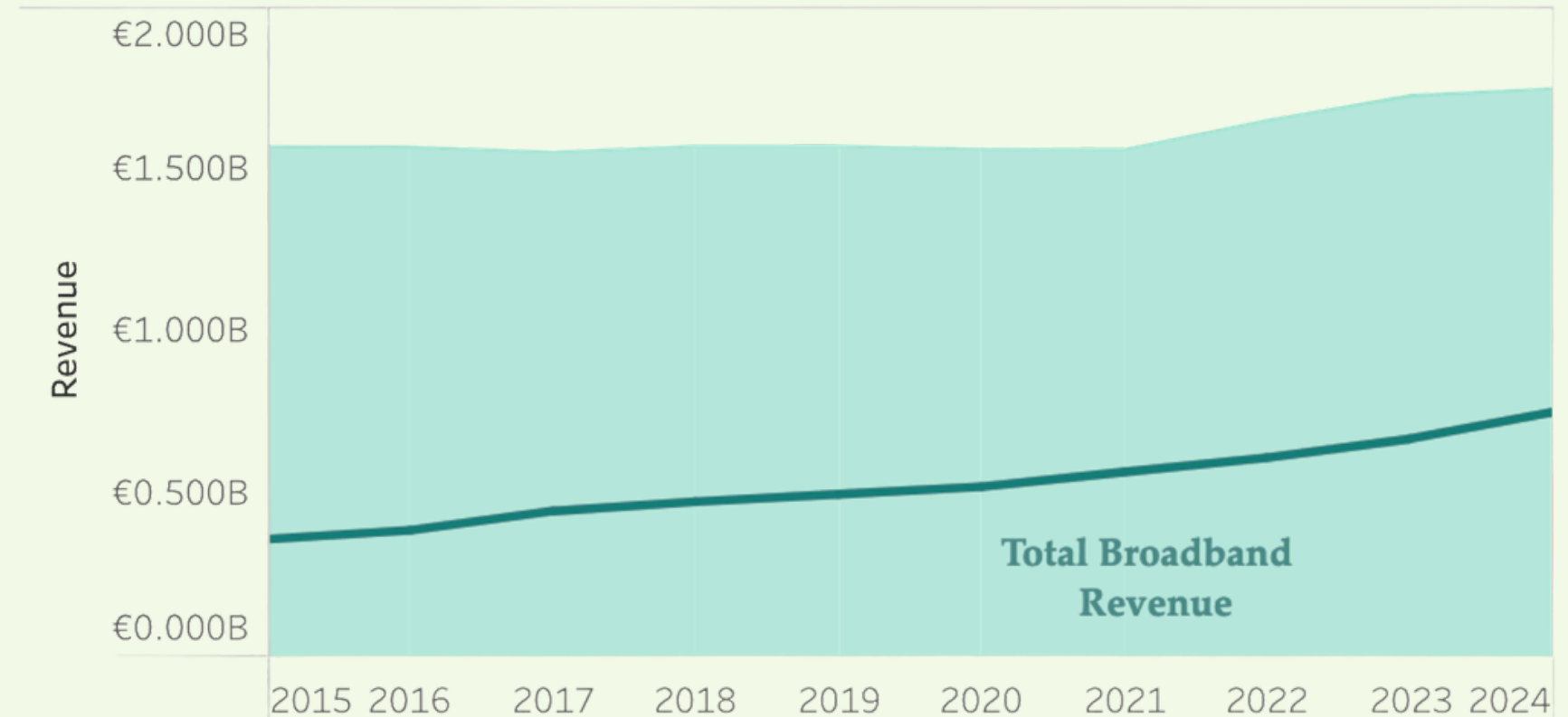
- * Revenue from Mobile Services stayed stable and saw a slight increase after 2021.

Growth is now driven by **subscriber volume**,
not individual customer value.

Mobile Data or Broadband?

- * Mobile Data dominates.
- * Broadband lags behind but sees a sharper increase (2x)
- * The total internet traffic channeled through broadband has seen a massive upwards trend.
- * Broadband dominates the data throughput.

Mobile vs Broadband Revenue



4k Streaming, Virtual Meetings, Video Calls, Remote Employment...

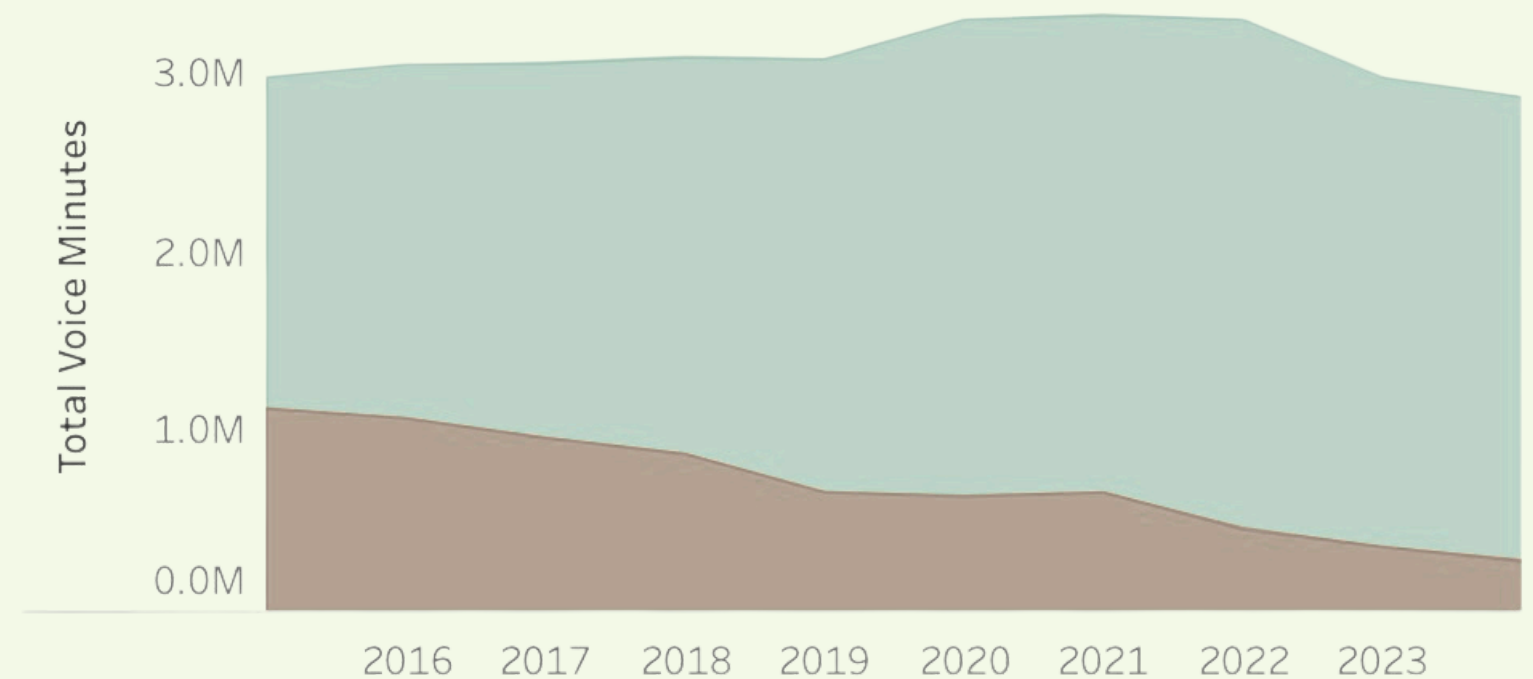
Can Mobile Data ever be reliable enough?

The Landline Demise



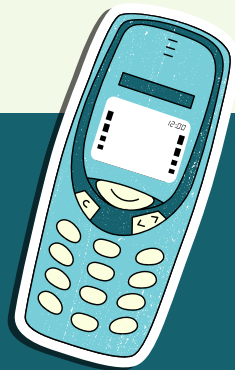
- 💡 Landline traffic has gradually **declined** over the decade.
- 💡 Mobile voice traffic has stayed steady yet **dominant**.
- 💡 Shift to **data-centric** modes of communication.
- 💡 **Rise** in data consumption and demand for internet.

Fixed Voice vs Mobile Traffic



Fixed Voice vs Mobile Traffic

- Fixed Voice Traffic (...)
- Mobile Traffic (mins)



Internet is slowly replacing **all other means** of communication

The Fiber Future



DSL Technology:

Went from being dominant to practically **non-existent**.

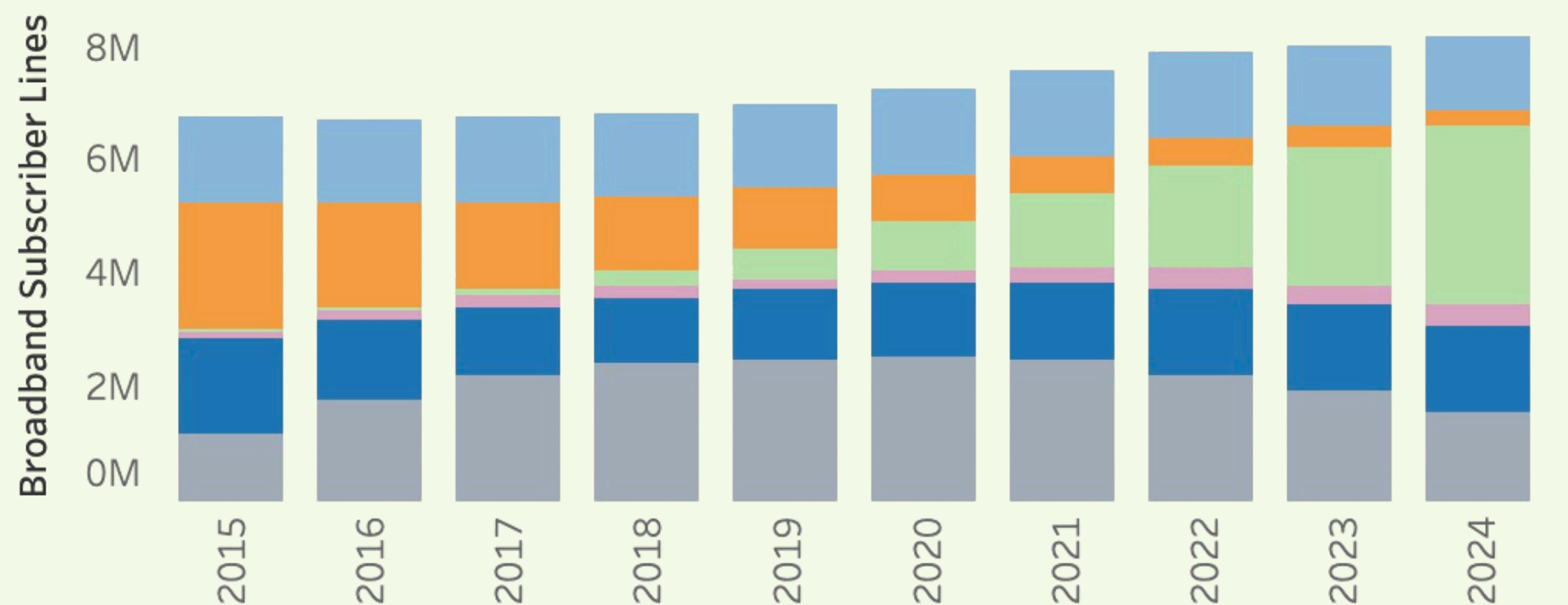
FTTP:

Usage **surged** owing to the faster speeds (up to 1-5 Gbps)

Mobile Data:

Steady usage with improvements in technology **(4G, 5G)**

Technology-wise Broadband Trends



The industry is betting its future on fiber infrastructure to meet data demand.

Subscriptions

Mobile Data:

Increase in subscribers is natural and **expected**.

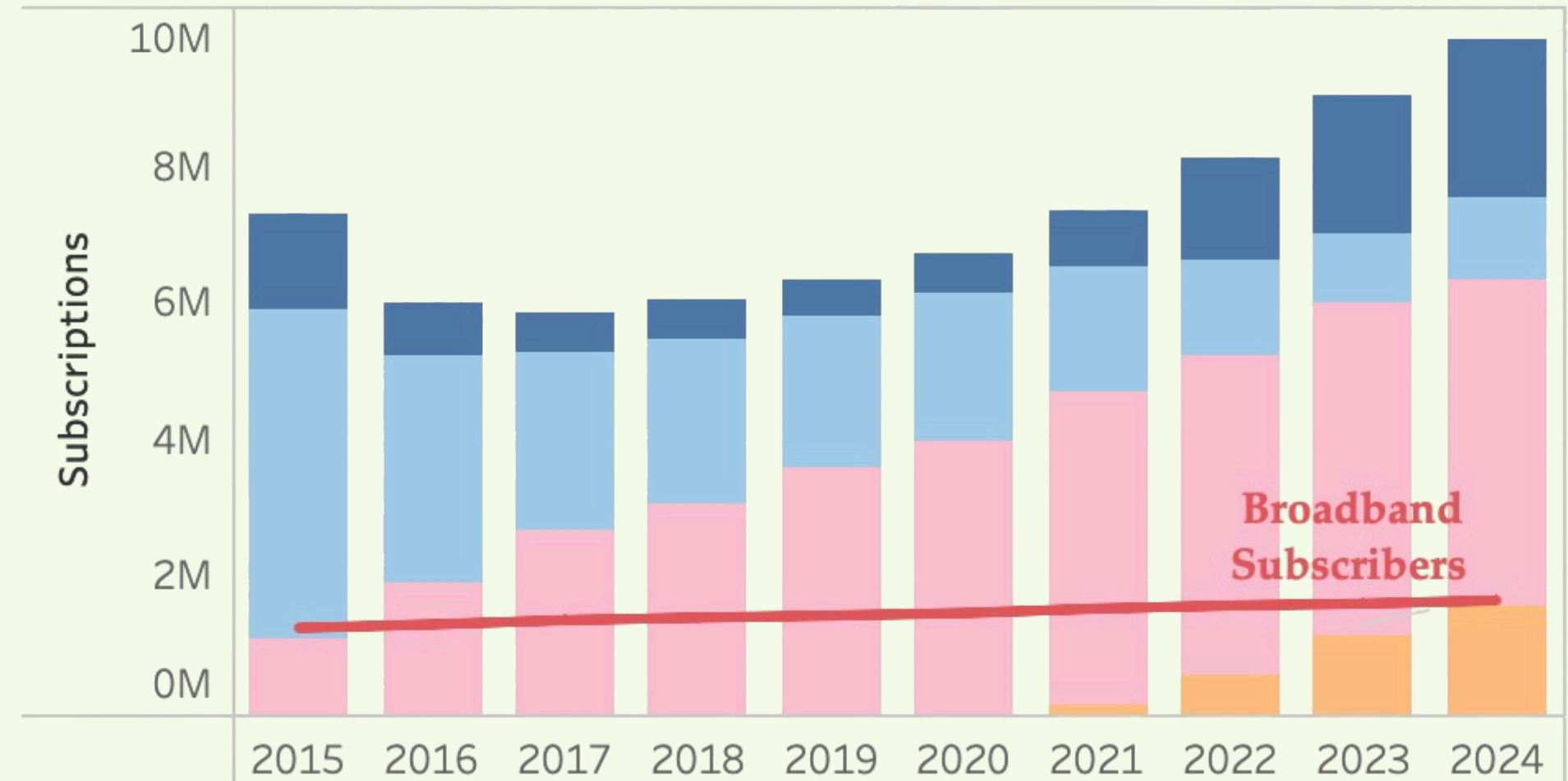
Broadband Data:

While the subscriber growth remains slow, the data consumption has increased **5 folds**.

Mobile Data

- 2G Subscriptions
- 3G Subscriptions
- 4G Subscriptions
- 5G Subscriptions

Broadband vs Mobile Data Usage



Subscriptions or traffic? What is the metric of success?

MARKET SHARE & INVESTMENT TRENDS

Mobile Market Share Stability

Vodafone, Three and Eir - maintain dominance.

Broadband Market Share

Eir, Vodafone and Virgin Mobile have all lost some market share but the outlook is stable.

Investment Trends

Broadband infrastructure investment is rising sharply nationwide.

Aligns with Ireland's digital strategy and EU connectivity goal.



Presented by Group 6

THANK YOU
